CSE 322 | AI
Class - 02 | 13-May-2026 | Shayan

Definition of AI

A way to make machines think & behave intelligently.

These machines are
controlled by SOFTWARE
↓
AI Softwares are
intelligent

AI is a science of
finding theories & methodologies

This helps machines
understand the world and
accordingly react to situations

AI Field

① Closely related to studying human ~~being~~ brain.
→ understanding how a brain works.

② Make machines that mimic the way
human learns thinks, & takes action.

③ Develop a platform to
→ develop intelligent systems
→ that are capable of learning.

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Studying AI

1. The field of AI tries to understand patterns & behaviors of entities.
2. Build smart systems and Understand the concept of AI. (Intelligence)
3. Construct useful systems.

How AI Works

Intelligence

↑

Understanding

↑ pattern extraction

Knowledge

↑ cognition

Information

↑ processing

Data

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Applications of AI

1. Computer Vision (CV)
2. Natural Language Processing (NLP)

Computer Vision:

1. Deals with visual data (images & videos)
2. Understand content
3. Extract insights

Natural Language Processing:

1. Deals with text content
2. Interaction with natural language
3. Usecase:
   → Search engine
   Perk: deliver right search results

* STT = Speech to Text
* TTS = Text to Speech

Common use of NLP:

1. Detecting malicious/harmful/hateful speech.

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↗ Applications of AI

③ Speech Recognition
④ Expert Systems

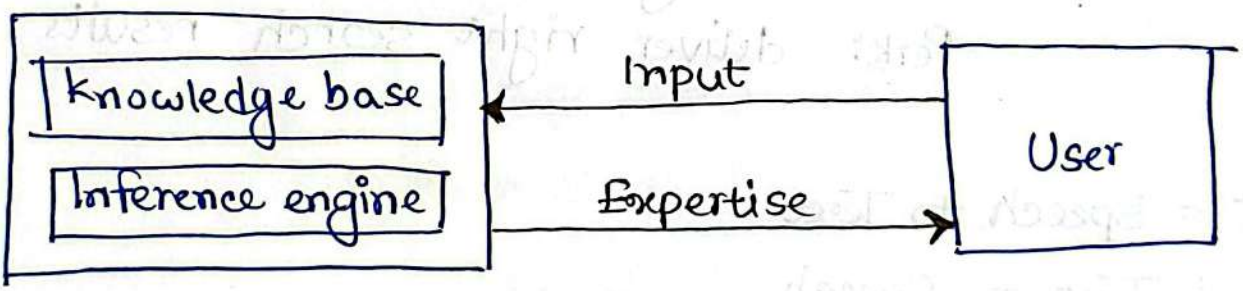
Speech Recognition:

① capable of hearing & understanding speech
② common example: ~~chat bots~~
- Smart Assistants

Expert Systems:

① Systems like this provide advice or make decisions.

② common fields:
- Finance
- Medicine
- Marketing



"if-condition"

→ Knowledge base = fact-based / conditions

→ data base = training content

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Applications of AI

5. Games
6. Robotics

Robotics :

1. combines many concepts of AI.
2. Consists of sensors & actuators.

Sensor → sensing → eye → 'seeing things'

vs

Actuator → taking actions → hand → measure:
- temperature
- heat
- movements

3. Capable of adapting to new environments.

Goals of AI

« self-study »

Branches of AI

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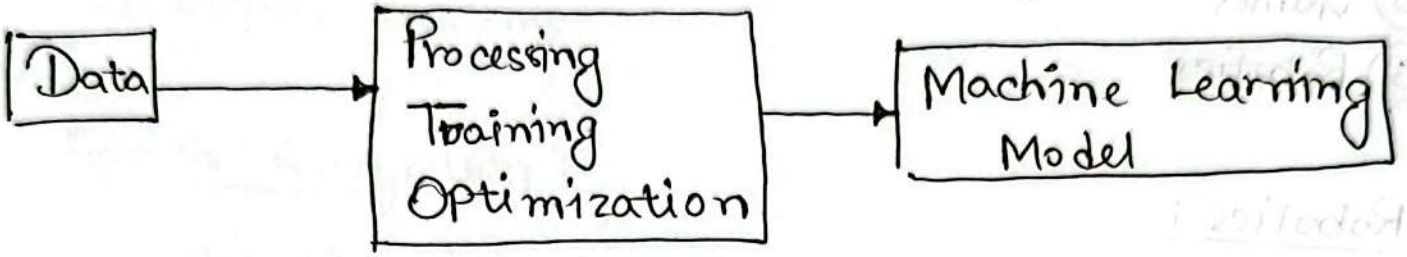
① Machine Learning & Pattern Recognition



- Learn from data
- Perform predictions on unknown data

② Logic-based AI

2.1: Mathematical logic

2.2: ~~Expects~~ Express facts & rules about a particular problem domain.

2.3: Searching
↳ a large number of possibilities are given, pick the most optimal one.

Semantic: Grammatical arrangement

Syntax: Logical arrangement

E.g.: Chess, Networking, Resource Allocation, Scheduling

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Branches of AI

③ Knowledge Representation

3.1 : Summerization

E.g. : Gemini

3.2 : (real world) → (mathematical possibilities)

Output : Google Assistant

④ Planning

Optimal planning that gives us maximum returns

Start with,

→ facts about particular situation
→ statement of a goal

Generate

→ most optimal plan to achieve the goal.

⑤ Heuristics

A heuristics is a technique used to solve a given problem that's practical and useful in solving the problem in the short-term but not guaranteed to be optimal. Fields : Robotics, Search Engine

We cannot check every single possibilities to pick the best option. E.g. : Indexing

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↗ Branches of AI

⑥ Genetic Programming

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Goals of AI

1. Systems that thinks like humans
2.
3.
4.

Goals of AI

1. <u>Thinking Humanity</u>: Cognitive Science
   - Effort to make computer think
   - Focus on reasoning process
   - How results are obtained

Goal(?)

- ☒ Produce human-like behavior.
- ☑ Produce a sequence of steps of the reasoning process, followed by a human in solving the same task.

Thinking Humanity	vs	Acting Humanly

exam question

2. <u>Acting Humanly: Turing Test</u>
   - Focus on actions
   - Get similar process of what a human being might do to obtain the answer.

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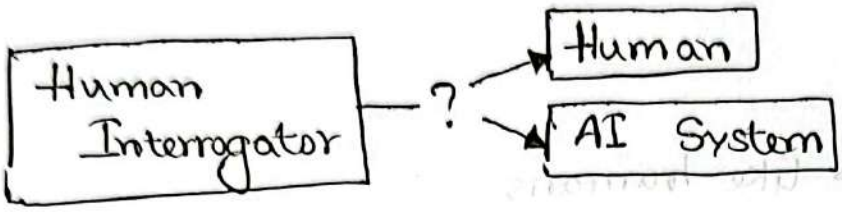
Turing Test

conducted in 1950



Turing's Predictions:

① By 2000s, A machine might have 30% chance of fooling a lay person for 5 minutes

② All major arguments against AI in following 50 years.

③ Thinking Rationally: Laws of Thought

"What is rational?"

① Possible to perceive, reason and act.

② Focus on inference mechanism,

how?

by focusing on things & techniques that are probably correct & guarantee an optimal solution

Goal is to formalize the reasoning process as a system of logical rules and procedures for inference.

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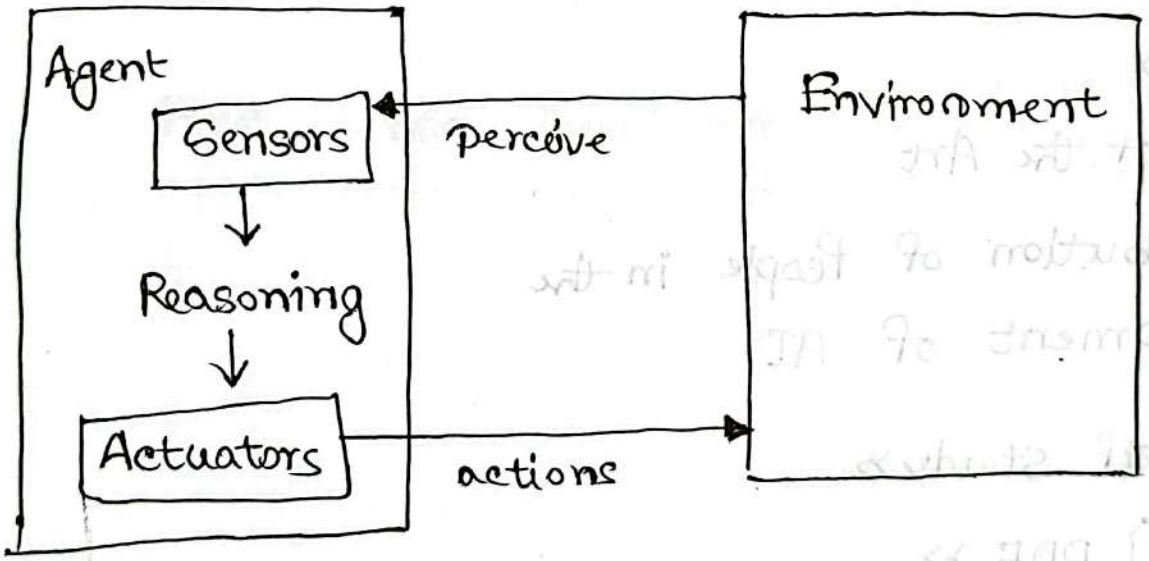
but,

'NOT ALL PROBLEM CAN BE SOLVED JUST BY REASONING & INFERENCE.'

Inference (অনুমান)
↳ learn from known data
↳ identify unknown data

Agent

Agent is anything that perceives the environment through sensors & acts upon it through actuators & effectors.



Building Rational Agent

"Rationality refers to doing the right thing in a given circumstances."

→ obtain maximum benefit to the entity performing the action.

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An agent is rational when,

→ given a set of rules, it takes actions to achieve its goals.

→ perceives & acts according to the information that is available.

↗ Building an Intelligent Agent

<< self study >>

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↗ Strong vs Weak AI

↗ History

↗ State of the Art

↗ Contribution of People in the development of AI

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Four Rules to be a better AI Agent

1. Must have the ability to perceive environment
2. Take decision by using perception
3. Decision should translate to actions
4. Provide rational & righ out-come.

Four Factors (Rationality of Agent)

1. ~~Rationality of agent~~ Performance → 'P
2. Environment → E
3. Actuators → A
4. Sensing → S'

Performance Measure

First sense, Then act
'S' 'A'